

Project Design Phase

Solution Architecture

SWTID-2026-3408

DATE	15 February 2025	MAXIMUM MARKS	4 Marks
TEAM ID	SWTID-2026-3408		
PROJECT NAME	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer		

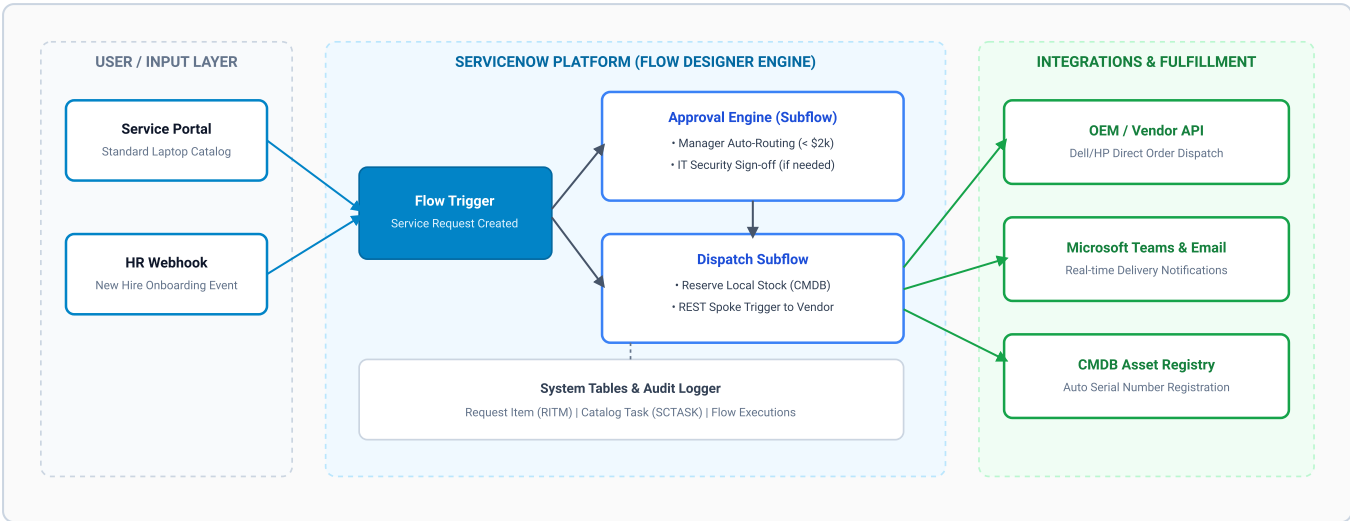
SOLUTION ARCHITECTURE GOALS & OVERVIEW

Solution architecture is a structural framework that bridges business requirements with technical execution to automate standard laptop orders using ServiceNow Flow Designer:

- Find the best tech solution to solve existing business problems.
- Describe structure, behavior, and data flow to stakeholders.
- Define automated workflows, triggers, and approval logic.
- Provide specifications for delivery and vendor integration.

SOLUTION ARCHITECTURE DIAGRAM

FIGURE 1: AUTOMATED LAPTOP PROCUREMENT FLOW ARCHITECTURE



Trigger Path A (Employee Self-Service): Direct request via ServiceNow Service Portal → Flow Designer executes manager approval → Order dispatched.

Trigger Path B (Automated Onboarding): HR System trigger via REST Webhook → Flow Designer generates pre-approved laptop order automatically.

CATALOG LAYER

Service Portal forms with dynamic variables, role-based laptop models, and pricing controls.

LOGIC ENGINE

ServiceNow Flow Designer with custom spokes, conditional branches, and subflows.

INTEGRATION LAYER

REST API Spokes connecting directly to external OEM vendor procurement portals.

ASSET TRACKING

CMDB lifecycle tables auto-populated with hardware serial numbers and user assignments.

Reference: ServiceNow Flow Designer Architecture & Integration Hub Spoke Design Standards (2025/2026).